Chunduri Navyatha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.03
Sri Chaitanya Junior College	Class XII (BIEAP)	2023	94.70%
Viswabharati High School	Class X (SSC)	2021	10.00%

EXPERIENCE

Research AI ML Intern (*Unistring Tech Solutions)*

- Designed a RAG based AI Agent for automated pytest test-case generation from input code functions, using AST-based chuncking with Tree-sitter to ensure syntactically complete, function level code chuncks.
- Evaluated Multiple LLMs on SWE-bench verified and HumanEval benchmarks and selected Qwen2.5-coder-32B-Instruct based on Benchmark performance.
- Built a hybrid code retrieval pipeline using CodeBert embeddings, Qdrant, BM25+Reciprocal Rank Fusion,NetworkX dependency graph traversal, and BGE cross encoder reranking.

PROJECTS

A Comparative Study of Load Balancing Algorithms

- Benchmarked 6 static(Round Robin, IP Hash) and dynamic(Least connection) load balancing algorithms by implementing a custom tree-topology network in Mininet to compare performance.
- Developed a multi-threaded Python system to simulate concurrent HTTP requests & analyzed performance.

Nika Hangouts:

- Built GenAI powered location intelligence chatbot using Vercel AI SDK and OpenAI for recommendation.
- Designed an interactive MapLibreJS interface with dynamic GeoJSON layer rendering for real time location visualization.
- Implemented LLM tool calling to integrate Nomination(OpenStreetMap) APIs for contextual geospatial search.
- Enabled secure API handling and streaming LLM responses to support real-time, low-latency chatbot interactions.

Wheat Head Detection using Deep Leaning(kaggle)

- Built and Benchmarked an object detection pipeline for wheat head detection on Kaggle's Global wheat dataset, comparing Faster R-CNN+VGG16, RT-DETR, and YOLO V26.
- Achieved 95.1% mAP@0.5 with YOLO at 19.03 FPS, 94.4% mAP@0.5 with RT-DETR at 15.78 FPS and 86% mAP0.5 with R-CNN+VGG16 at 2.75 FPS.
- Found YOLO and RT-DETR performed comparably on accuracy, with YOLO offering a slight edge in both mAP and inference speed, while both significantly outperformed faster R-CNN+VGG16.

AI Email Summariser Agent

- Developed an AI agent to fetch, classify, and summarize unread emails, automating inbox triage with concise AI-generated summaries.
- Authored technical documentation covering system architecture and evaluation logs for the project.

SKILLS

- **Languages:** C++, C, Python, SQL,JavaScript, HTML/CSS,PostgreSQL.
- **Tools & Technologies:** GitHub, Mininet, LaTeX.
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-Learn, NumPy, pandas, matplotlib, seaborn.

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Operating Systems-1, Database Management-1, Computer Networks, Theory of Computation.

- **AI & Mathematics:** Machine Learning, Probability, Statistics, Linear Algebra, Discrete Maths

POSITIONS OF RESPONSIBILITY

Sponsorship Coordinator -Elan & nVision ,IIT Hyderabad:

- Worked as a sponsorship coordinator for Elan & nVision, reaching out to 20+ companies as a part of team to secure funding for the college fest.
- Managed all sponsor deliverables end-end , from proposals to on-ground execution.
- Served as a key point of contact between sponsors and organizing committee, ensuring timely fulfillment of commitments.

Sunshine Mentor-Mentored Junior students through structured group sessions, guiding their academic and personal transition into college.

Member, Block chain Club- associated with orientation activities of the club for freshers.

ACHIEVEMENTS

- **Semi Finalist, Flipkart GRID 7.0 & GRID 8.0-** Advanced through multiple competitive elimination rounds, selected from thousands of paticipants nationwide across engineering and ML tracks.
- **Reliance Foundation Scholar-**1 of 5000 elected nationlly through rigorous nationwide examination from lakhs of applicants, awarded or outstanding academic merit.
- selected for Google's upskilling program covering Data Structures & Algorithms and technical interview preparation.